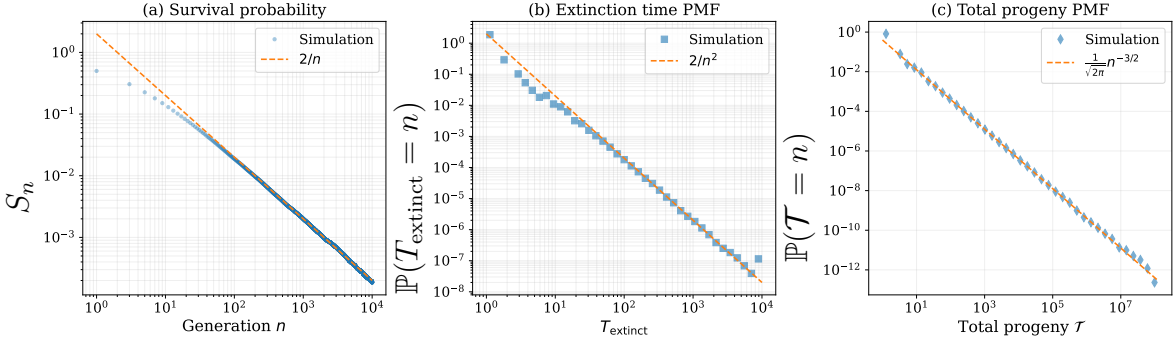

(a) Survival probability
Simulation
2/n
S_n
Generation n
(b) Extinction time PMF
Simulation
$2/n^2$
$\mathbb{P}(T_{\text{extinct}} = n)$
T_{extinct}
(c) Total progeny PMF
Simulation
$\frac{1}{\sqrt{2\pi}} n^{-3/2}$
$\mathbb{P}(\mathcal{T} = n)$
Total progeny $\mathcal{T}$